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CUET UG

Mathematics

Integration and its Applications

Indefinite & Definite Integrals, Properties, Leibniz Rule, and Area Bounded by Curves

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Chapter Overview: The Integral Calculus Engine

Integration represents one of the dual pillars of infinitesimal calculus, serving both as the inverse operation to differentiation (the **antiderivative**) and as the rigorous foundational mechanism for computing continuous accumulations, total displacements, and geometrically bounded areas. In the CUET UG Mathematics examination, Integral Calculus constitutes between **18% to 22% of total exam weightage**, spanning both Section A (compulsory common core) and Section B1 (core mathematics).

A mastery of integration demands far more than mechanical memorization of antiderivative tables. High-scoring aspirants must master three interconnected intellectual capabilities:

1. **Pattern Recognition in Indefinite Integrals:** Instantly diagnosing whether an integrand yields to direct algebraic manipulation, trigonometric substitution, integration by parts via the ILATE hierarchy, partial fraction decomposition, or one of the 9 special quadratic forms.
2. **Definite Integral Symmetries & The King's Rule:** Utilizing definite integral properties (P_0 through P_7) to collapse intractable integrands into trivial constants without ever finding the indefinite antiderivative.
3. **Geometric Precision in Area Bounded (AOI):** Accurately sketching bounding curves, locating intersections, choosing the optimal axis of slicing (vertical strips dx vs horizontal strips dy), and deploying high-speed competitive shortcuts for parabolas, ellipses, and trigonometric loops.

THE FUNDAMENTAL DUALITY OF CALCULUS

Let $f : [a, b] \to \mathbb{R}$ be a continuous real-valued function:

- **The Indefinite Integral:** $\int f(x)dx = F(x) + C$, where $F'(x) = f(x)$ and $C \in \mathbb{R}$ is an arbitrary constant of integration representing an entire one-parameter family of mutually parallel curves.
- **The First Fundamental Theorem of Calculus (FTC 1):** If f is continuous on $[a, b]$ and $A(x) = \int_a^x f(t)dt$, then $A'(x) = f(x)$ for all $x \in [a, b]$.
- **The Second Fundamental Theorem of Calculus (FTC 2):** If $F'(x) = f(x)$ on $[a, b]$, then $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$. The arbitrary constant C identically cancels out.

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1. Indefinite Integration: Methods & Core Forms

An indefinite integral represents the totality of all antiderivatives of a given function. If $F(x)$ satisfies $\frac{d}{dx}[F(x)] = f(x)$, then:

$$\int f(x)dx = F(x) + C$$

Geometrically, $y = F(x) + C$ represents a family of congruent curves obtained by translating the graph of $y = F(x)$ vertically along the y-axis. The tangents to all these curves at any ordinate $x = x_0$ are strictly parallel, each possessing identical slope $m = f(x_0)$.

1. Standard Elementary Antiderivatives

Integrand Family	Indefinite Integral Formula	Domain Restrictions / Notes
Power Functions	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$	$n \neq -1$. Add $+1$ to power.

Integrand Family	Indefinite Integral Formula	Domain Restrictions / Notes
Reciprocal Function	$\int \frac{1}{x}dx = \ln\|x\| + C$	$x \neq 0$. Absolute value mandatory.
Natural Exponential	$\int e^{kx}dx = \frac{e^{kx}}{k} + C$	$k \neq 0$. Linear argument scaling.
General Exponential	$\int a^x dx = \frac{a^x}{\ln a} + C$	$a > 0, a \neq 1$. Divide by $\ln a$.
Sine Function	$\int \sin(kx)dx = -\frac{\cos(kx)}{k} + C$	Note the negative sign!
Cosine Function	$\int \cos(kx)dx = \frac{\sin(kx)}{k} + C$	Standard positive antiderivative.
Secant Squared	$\int \sec^2 x dx = \tan x + C$	From $\frac{d}{dx}[\tan x] = \sec^2 x$.
Cosecant Squared	$\int \csc^2 x dx = -\cot x + C$	From $\frac{d}{dx}[\cot x] = -\csc^2 x$.
Secant Tangent	$\int \sec x \tan x dx = \sec x + C$	Direct trigonometric derivative.
Cosecant Cotangent	$\int \csc x \cot x dx = -\csc x + C$	Negative trigonometric derivative.
Tangent Function	$\int \tan x dx = \ln\|\sec x\| + C = -\ln\|\cos x\| + C$	Derived via $\sin \frac{x}{\cos} x$ substitution.
Cotangent Function	$\int \cot x dx = \ln\|\sin x\| + C$	Derived via $\cos \frac{x}{\sin} x$ substitution.
Secant Function	$\int \sec x dx = \ln\|\sec x + \tan x\| + C$	Alternative: $\ln\|\tan(\frac{\pi}{4} + \frac{x}{2})\| + C$.
Cosecant Function	$\int \csc x dx = \ln\|\csc x - \cot x\| + C$	Alternative: $\ln\|\tan(\frac{x}{2})\| + C$.

2. Integration by Substitution (Change of Variable)

When an integrand cannot be integrated directly in its initial variable x, we transform the variable to t using an invertible substitution $x = g(t)$, giving $dx = g'(t)dt$:

$$\int f(g(x))g'(x)dx = \int f(t)dt \quad \text{where } t = g(x)$$

High-Yield Substitution Archetypes in CUET:

1. **The Golden Derivative-over-Function Form:**

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$$

Example: $\int \frac{2x+3}{x^2+3x+7}dx = \ln|x^2 + 3x + 7| + C$.

2. **The Power Function Derivative Form:**

$$\int [f(x)]^n f'(x)dx = \frac{[f(x)]^{n+1}}{n+1} + C \quad (n \neq -1)$$

Example: $\int \sin^4 x \cos x dx = \frac{\sin^5 x}{5} + C$.

3. **Trigonometric Integrands with Odd/Even Powers:**
   - If power of $\sin x$ is odd, substitute $t = \cos x$.
   - If power of $\cos x$ is odd, substitute $t = \sin x$.
   - If powers of both $\sin x$ and $\cos x$ are even, convert using half-angle identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

STANDARD ALGEBRAIC & TRIGONOMETRIC SUBSTITUTIONS

When encountering algebraic radical expressions in the integrand:

- For $\sqrt{a^2 - x^2}$: Substitute $x = a\sin\theta$ or $x = a\cos\theta$ (since $1 - \sin^2\theta = \cos^2\theta$).
- For $\sqrt{a^2 + x^2}$ or $a^2 + x^2$: Substitute $x = a\tan\theta$ or $x = a\sinh\theta$ (since $1 + \tan^2\theta = \sec^2\theta$).
- For $\sqrt{x^2 - a^2}$: Substitute $x = a\sec\theta$ (since $\sec^2\theta - 1 = \tan^2\theta$).
- For $\sqrt{\frac{a-x}{a+x}}$: Substitute $x = a\cos 2\theta$.

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2. The Special 9 Quadratic Integrals

Integrals involving quadratic expressions in the denominator or numerator form a cornerstone of the CUET syllabus. Mastering their exact functional forms and avoiding common coefficient errors is mandatory.

The Three Non-Radical Rational Denominators

1. $$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$$
2. $$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$
3. $$\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$$

The Three Radical Denominators

4. $$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C \quad \left(\textbf{CRITICAL: No } \frac{1}{a} \textbf{ factor outside!}\right)$$
5. $$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln|x + \sqrt{x^2 - a^2}| + C$$
6. $$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln|x + \sqrt{x^2 + a^2}| + C$$

The Three Radical Numerators (The AOI Triad)

7. $$\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$$
8. $$\int \sqrt{x^2 - a^2}\,dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ln|x + \sqrt{x^2 - a^2}| + C$$
9. $$\int \sqrt{x^2 + a^2}\,dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ln|x + \sqrt{x^2 + a^2}| + C$$

LINEAR OVER QUADRATIC DECOMPOSITION ALGORITHM

To evaluate integrals of the form $\int \frac{px+q}{ax^2+bx+c}dx$ or $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$:

1. Express the numerator in terms of the derivative of the quadratic radicand:

$$px + q = A\frac{d}{dx}[ax^2 + bx + c] + B = A(2ax + b) + B$$

2. Equate coefficients of x and constants to solve for A and B:

$$A = \frac{p}{2a}, \quad B = q - \frac{pb}{2a}$$

3. Split the integral into two distinct parts:

$$\int \frac{px+q}{ax^2+bx+c}dx = A\underbrace{\int \frac{2ax+b}{ax^2+bx+c}dx}_{\ln|\text{quadratic}|} + B\underbrace{\int \frac{dx}{ax^2+bx+c}dx}_{\text{Complete the square}}$$

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3. Integration by Parts & Partial Fractions

1. Integration by Parts & The ILATE Protocol

Derived directly from the product rule of differentiation $\frac{d}{dx}[uv] = uv' + vu'$, the formula for integration by parts is:

$$\int u(x)v'(x)dx = u(x)v(x) - \int u'(x)v(x)dx$$

Or in standard student shorthand:

$$\int (\textbf{First}) \cdot (\textbf{Second})dx = (\textbf{First})\int (\textbf{Second})dx - \int \left[\frac{d}{dx}(\textbf{First}) \cdot \left(\int (\textbf{Second})dx\right)\right]dx$$

THE ILATE PRIORITY HIERARCHY

To select which factor serves as the **First Function $u(x)$** (the function to differentiate) versus the **Second Function $v'(x)$** (the function to integrate), follow the strict priority order:

- **I — Inverse Trigonometric Functions:** $\sin^{-1}x, \cos^{-1}x, \tan^{-1}x$ (Highest priority for u)
- **L — Logarithmic Functions:** $\ln x, \log_a x$
- **A — Algebraic Functions:** $x^n, x^2+1, \sqrt{x}$
- **T — Trigonometric Functions:** $\sin x, \cos x, \sec^2 x$
- **E — Exponential Functions:** e^x, a^x (Lowest priority for u, ideal second function)

Crucial Rule: When integrating isolated inverse trigonometric or logarithmic functions (e.g. $\int \ln x dx$ or $\int \tan^{-1}x dx$), introduce unity $1 = x^0$ as the algebraic second function!

$$\int \ln x dx = \int \underbrace{(\ln x)}_{u} \cdot \underbrace{1}_{v'}dx = (\ln x)x - \int \left(\frac{1}{x} \cdot x\right)dx = x\ln x - x + C = x(\ln x - 1) + C$$

2. The Exponential Master Shortcut: $\int e^x[f(x) + f'(x)]dx$

A standard, exceptionally frequent question type in CUET involves the product of an exponential function and a sum of terms:

$$\int e^x[f(x) + f'(x)]dx = e^x f(x) + C$$

Verification: By the product rule, $\frac{d}{dx}[e^x f(x)] = e^x f(x) + e^x f'(x) = e^x[f(x) + f'(x)]$.

- **Classic Example 1:** $\int e^x [\tan x + \sec^2 x] dx = e^x \tan x + C$, since $f(x) = \tan x \implies f'(x) = \sec^2 x$.
- **Classic Example 2:** $\int e^x \left[\frac{x-1}{x^2}\right] dx = \int e^x \left[\frac{1}{x} - \frac{1}{x^2}\right] dx = \frac{e^x}{x} + C$, since $f(x) = \frac{1}{x} \implies f'(x) = -\frac{1}{x^2}$.
- **General Form:** $\int e^{kx} [k f(x) + f'(x)] dx = e^{kx} f(x) + C$.

3. Rational Functions and Partial Fractions

When integrating a rational function $\frac{P(x)}{Q(x)}$:

1. **Proper Fraction Verification:** Verify that $\deg(P) < \deg(Q)$. If $\deg(P) \geq \deg(Q)$, perform polynomial long division first to express $\frac{P(x)}{Q(x)} = T(x) + \frac{R(x)}{Q(x)}$ where $\deg(R) < \deg(Q)$.
2. **Linear Non-Repeated Factors:**

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

   Cover-Up Method: $A = \left[\frac{px+q}{x-b}\right]_{x=a}$ and $B = \left[\frac{px+q}{x-a}\right]_{x=b}$.
3. **Repeated Linear Factors:**

$$\frac{px^2+qx+r}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$$

4. **Irreducible Quadratic Factor:**

$$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

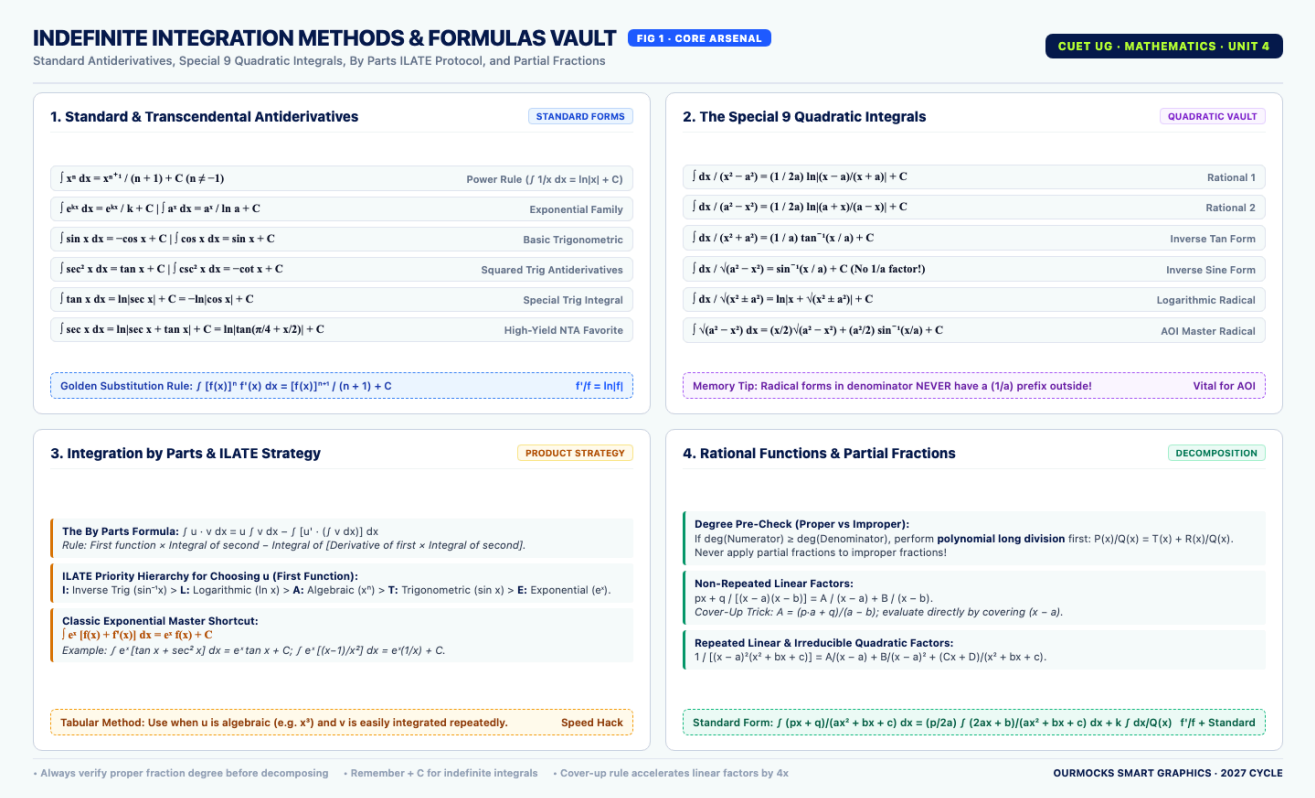


Figure 1: Figure 1: Comprehensive Indefinite Integration Methods, Special 9 Forms, ILATE Strategy, and Partial Fraction Decompositions.

Key Analytical Takeaways from Figure 1:

1. **The Three Radical Denominators Contrast:** Observe the critical structural difference between $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$ (which has NO $\frac{1}{a}$ factor outside) versus $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$ (which possesses a mandatory $\frac{1}{a}$ prefix). NTA frequently inserts answer options containing an erroneous $\frac{1}{a}$ prefix on inverse sine integrals.

2. **Proper Fraction Prerequisite:** Attempting partial fractions on improper fractions without long division produces fatal algebraic contradictions.
3. **Cover-Up Acceleration:** For linear factors, the Heaviside cover-up method computes partial fraction coefficients in under 5 seconds, bypassing simultaneous equations.

COVER-UP METHOD VS SIMULTANEOUS LINEAR EQUATIONS

Problem: Evaluate the indefinite integral $\int \frac{2x+1}{(x-1)(x+2)(x-3)}dx$.

- **Standard Decomposition:** $\frac{2x+1}{(x-1)(x+2)(x-3)} = \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x-3}$.
- **Heaviside Cover-Up Speed Technique:**
  - For A (set $x = 1$, cover $(x-1)$): $A = \frac{2(1)+1}{(1+2)(1-3)} = \frac{3}{(3)(-2)} = -\frac{1}{2}$.
  - For B (set $x = -2$, cover $(x+2)$): $B = \frac{2(-2)+1}{(-2-1)(-2-3)} = \frac{-3}{(-3)(-5)} = -\frac{1}{5}$.
  - For C (set $x = 3$, cover $(x-3)$): $C = \frac{2(3)+1}{(3-1)(3+2)} = \frac{7}{(2)(5)} = \frac{7}{10}$.
- **Antiderivative:**

$$\int \frac{2x+1}{(x-1)(x+2)(x-3)}dx = -\frac{1}{2}\ln|x-1| - \frac{1}{5}\ln|x+2| + \frac{7}{10}\ln|x-3| + C$$

CUET Speed Impact: Solves the partial fraction decomposition in 20 seconds flat without setting up or solving a 3×3 linear system!

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4. Definite Integrals: The Master Properties (P_0 to P_7)

A definite integral computes the net signed accumulation of a continuous function across a closed interval:

$$\int_a^b f(x)dx = \lim_{\{n\to\infty\}} h \sum_{\{r=1\}}^{n} f(a+rh) \quad \text{where } h = \frac{b-a}{n}$$

In CUET examination problems, explicit calculation of limits of Riemann sums is rarely required; instead, the vast majority of problems test deep mastery of the **8 Fundamental Definite Integral Properties**.

The 8 Canonical Properties of Definite Integrals

Property	Mathematical Statement	Exam Application / Utility
P_0	$\int_a^b f(x)dx = \int_a^b f(t)dt$	Dummy variable invariance.
P_1	$\int_a^b f(x)dx = -\int_b^a f(x)dx$	Limit interchange negates the integral; $\int_a^a = 0$.
P_2	$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$	Essential for modulus $\lvert x-c\rvert$ & step $[x]$ functions.
P_3	$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$	General King's Property: Reflection across midpoint.
P_4	$\int_0^a f(x)dx = \int_0^a f(a-x)dx$	Standard King's Property: The ultimate CUET weapon.
P_5	$\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$	Interval halving decomposition.
P_6	$\int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx & \text{if } f(2a-x)=f(x) \\ 0 & \text{if } f(2a-x)=-f(x) \end{cases}$	Queen's Property: Slices upper limit in half.
P_7	$\int_{-a}^a f(x)dx = \begin{cases} 2\int_0^a f(x)dx & \text{if } f(-x)=f(x) \quad \textbf{(Even)} \\ 0 & \text{if } f(-x)=-f(x) \quad \textbf{(Odd)} \end{cases}$	Symmetric limits: Odd functions vanish identically!

Deep Dive: The King's Property (P_4) Algorithm

The King's Property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ is responsible for solving over 60% of all definite integral problems in competitive examinations.

THE KING'S PROPERTY SYMMETRY COLLAPSE ALGORITHM

Whenever an integrand contains complementary trigonometric or exponential pairs on $[0,a]$ (such as $\sin x$ and $\cos x$ on $[0,\frac{\pi}{2}]$, or $f(x)$ and $f(a-x)$):

1. Label the original integral:
$$I = \int_0^a \frac{g(x)}{g(x)+g(a-x)}dx$$
2. Apply Property P_4 ($x \to a-x$):
$$I = \int_0^a \frac{g(a-x)}{g(a-x)+g(x)}dx$$
3. Add the two equations ($I + I = 2I$):
$$2I = \int_0^a \frac{g(x)+g(a-x)}{g(x)+g(a-x)}dx = \int_0^a 1dx = [x]_0^a = a$$

4. Solve for I:

$$I = \frac{a}{2}$$

Standard Classical Results via King's Rule:

- $\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}$ for ANY real exponent n.
- $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx = \frac{\pi}{4}$.
- $\int_0^{\frac{\pi}{2}} \ln(\sin x) dx = \int_0^{\frac{\pi}{2}} \ln(\cos x) dx = -\frac{\pi}{2} \ln 2$.

The Leibniz Rule of Differentiation Under the Integral Sign

When evaluating limits of indeterminate forms ($\frac{0}{0}$ or $\frac{\infty}{\infty}$) where integrals appear in the numerator, L'Hôpital's Rule requires differentiating the integral. The Leibniz Rule gives:

$$\frac{d}{dx}\left[\int_{u(x)}^{v(x)} f(t)dt\right] = f(v(x)) \cdot v'(x) - f(u(x)) \cdot u'(x)$$

Example: To find $\lim_{x\to 0} \frac{\int_0^{x^2} \cos(t^2)dt}{x \sin x}$:

- Direct substitution yields $\frac{0}{0}$. Applying L'Hôpital's Rule:

$$\text{Numerator derivative} = \cos\left((x^2)^2\right) \cdot \frac{d}{dx}[x^2] - 0 = 2x\cos(x^4)$$

$$\text{Denominator derivative} = \frac{d}{dx}[x \sin x] = \sin x + x\cos x$$

- Taking the limit:

$$\lim_{x\to 0} \frac{2x\cos(x^4)}{\sin x + x\cos x} = \lim_{x\to 0} \frac{2\cos(x^4)}{\frac{\sin x}{x} + \cos x} = \frac{2(1)}{1+1} = 1$$

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5. Applications of Integrals: Bounded Area (AOI)

The Application of Integrals (AOI) in CUET tests geometric comprehension, spatial visualization of curves, and precise definite integration to evaluate the surface area of regions bounded by lines, parabolas, circles, ellipses, and trigonometric functions.

1. Fundamental Theorems of Bounded Area

1. **Area Bounded by $y = f(x)$, the x-axis, and Ordinates $x = a, x = b$:**

$$\text{Area} = \int_a^b |y|\, dx = \int_a^b |f(x)|\, dx$$

If the curve crosses the x-axis at $x = c \in (a, b)$, such that $f(x) \geq 0$ on $[a, c]$ and $f(x) \leq 0$ on $[c, b]$:

$$\text{Area} = \int_a^c f(x)dx + \left|\int_c^b f(x)dx\right| = \int_a^c f(x)dx - \int_c^b f(x)dx$$

Warning: Never compute $\int_a^b f(x)dx$ without modulus splitting! Net signed area allows negative lobes to cancel positive lobes, yielding an erroneous total area (e.g. $\int_0^{2\pi} \sin x dx = 0$, whereas the bounded area is 4 sq units).

2. **Area Bounded by $x = g(y)$, the y-axis, and Abscissae $y = c, y = d$:**

$$\text{Area} = \int_c^d |x| \, dy = \int_c^d |g(y)| \, dy$$

Horizontal slicing is vastly superior when curves are explicitly solved for x in terms of y (e.g. $x = y^2$).

3. **Area Between Two Curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$:**

$$\text{Area} = \int_a^b \left[y_{\text{upper}} - y_{\text{lower}} \right] dx = \int_a^b [f(x) - g(x)] dx \quad \text{where } f(x) \geq g(x)$$

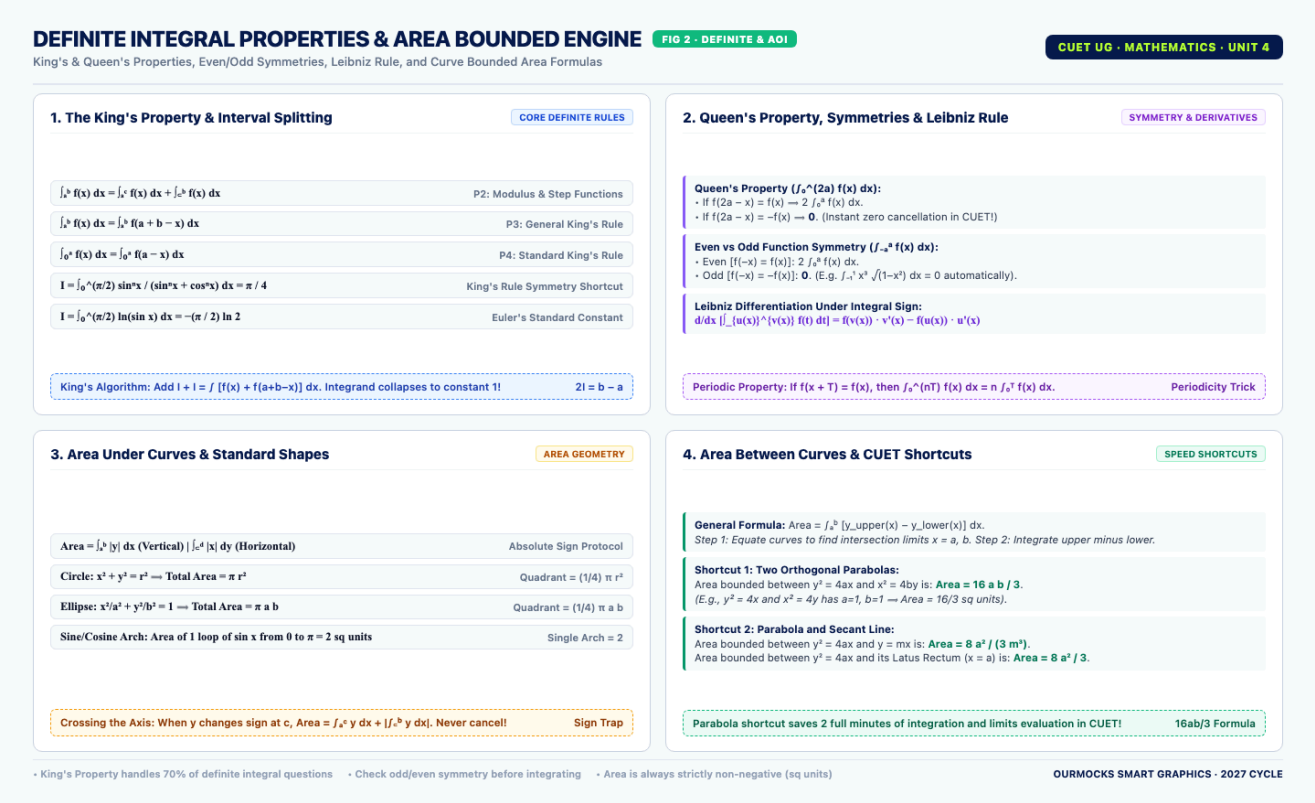


Figure 2: Figure 2: Definite Integral Properties, Symmetry & Leibniz Rule, and Area Bounded by Curves Geometric Engine.

Key Analytical Takeaways from Figure 2:

1. **The Modulus Area Integrity Principle:** Area is a strictly non-negative physical quantity. When a curve dips below the x-axis, the negative definite integral must be negated to yield positive area.
2. **Orthogonal Parabola 16ab/3 Rule:** The region enclosed between two standard mutually orthogonal parabolas $y^2 = 4ax$ and $x^2 = 4by$ is always equal to $\frac{16ab}{3}$ sq units. For $y^2 = 4x$ and $x^2 = 4y$, $a = 1, b = 1 \implies \text{Area} = \frac{16}{3}$.
3. **Parabola & Line Shortcut:** The area enclosed by the parabola $y^2 = 4ax$ and the line $y = mx$ through the origin is given by $\frac{8a^2}{3m^3}$.

6. High-Speed CUET Shortcuts for Area Calculations

To finish the 50-question CUET Mathematics paper within 60 minutes, aspirants must deploy proven analytical shortcuts for standard conic sections and closed regions:

Geometric Region	Standard Equation(s)	Direct Area Formula (sq units)
Standard Circle	$x^2 + y^2 = r^2$	πr^2 (First quadrant $= \frac{1}{4}\pi r^2$)
Standard Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	πab (First quadrant $= \frac{1}{4}\pi ab$)
Parabola & Latus Rectum	$y^2 = 4ax$ and $x = a$	$\frac{8}{3}a^2$
Two Orthogonal Parabolas	$y^2 = 4ax$ and $x^2 = 4by$	$\frac{16}{3}ab$
Parabola & Secant Line	$y^2 = 4ax$ and $y = mx$	$\frac{8a^2}{3m^3}$
Single Loop of Sine/Cosine	$y = \sin x, x \in [0, \pi]$ or $y = \cos x, x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$	2 sq units
Between $y = \sin x$ and $y = \cos x$	$x \in [\frac{\pi}{4}, 5\frac{\pi}{4}]$	$2\sqrt{2}$ sq units
Parabola & Double Ordinate	$y^2 = 4ax$ from $x = 0$ to $x = x_0$	$\frac{4}{3}x_0y_0 =$ $\frac{2}{3}\cdot$ (Bounding Rectangle)

ARCHIMEDES' QUADRATURE OF THE PARABOLA

Archimedes' classical theorem proves that the area bounded by any parabolic arc and a chord is exactly **two-thirds ($\frac{2}{3}$)** of the area of the circumscribed parallelogram/rectangle sharing the same base and height:

$$\text{Area} = \frac{2}{3} \cdot (\text{Base}) \cdot (\text{Height})$$

For $y^2 = 4ax$ bounded by $x = a$ (latus rectum):

- Base (latus rectum length) $= 4a$. Height $= a$.
- Area $= \frac{2}{3} \cdot (4a) \cdot (a) = \frac{8a^2}{3}$ sq units!

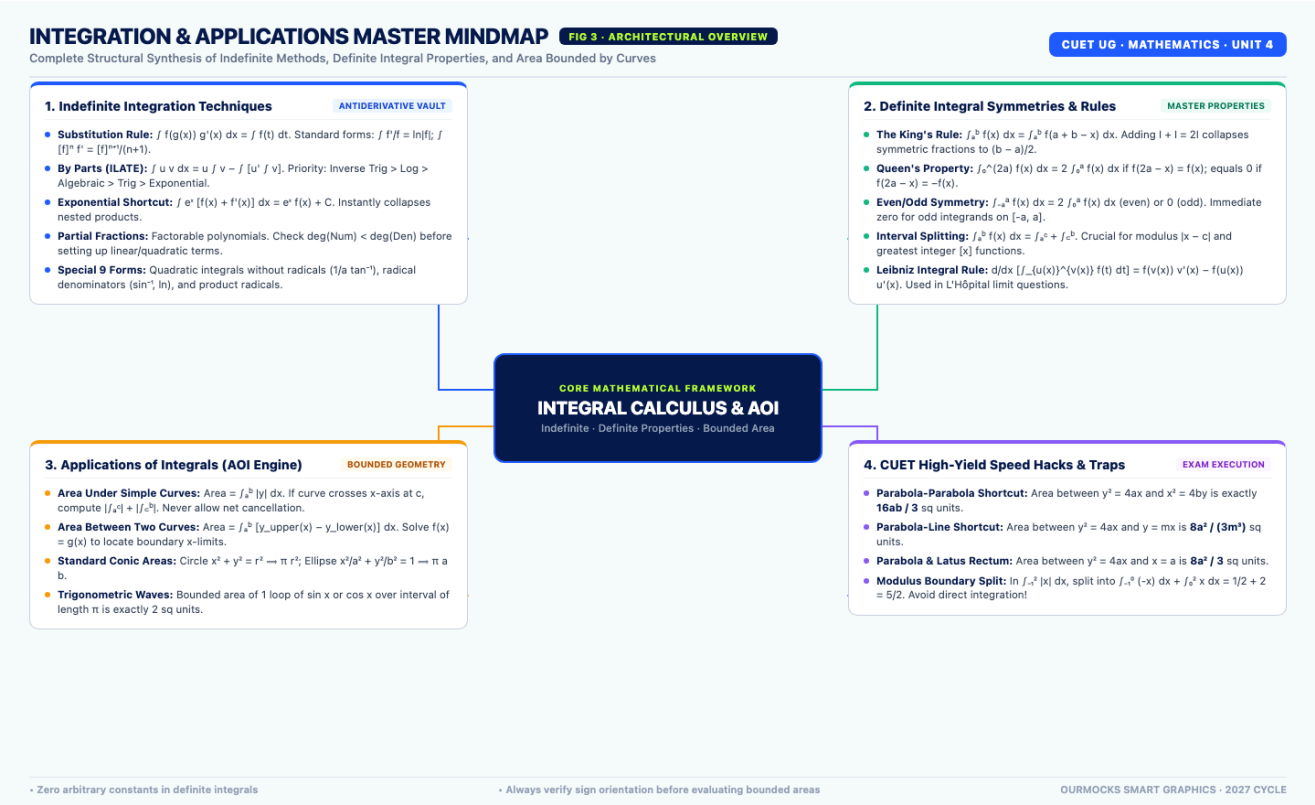


Figure 3: Figure 3: Integral Calculus Architectural Mindmap — Synthesis of Indefinite Methods, Definite Integral Properties, and Area Bounded by Curves.

Key Architectural Takeaways from Figure 3:

1. **Unified Calculus Hierarchy:** Integral calculus flows seamlessly from local differential inversions (antiderivatives) to global accumulative invariants (definite integral properties) and spatial physical measurements (area bounded).
2. **Strategic Prerequisite Flow:** Always test for symmetry (P_7 odd/even or P_4 King's Rule) BEFORE attempting algebraic manipulation on definite integrals.
3. **Algebraic Sanity in AOI:** Solve for intersection points first. The limits of integration $x = a, b$ are ALWAYS the real roots of $f(x) - g(x) = 0$.

7. High-Yield Revision Cheatsheet

Concept / Formula	Mathematical Expression	Exam Anchor
Power Rule	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$	Algebraic Base
Logarithmic Integral	$\int \frac{1}{x} dx = \ln\|x\| + C$	Reciprocal Form
Exponential Form	$\int a^x dx = \frac{a^x}{\ln a} + C; \quad \int e^{kx} dx = \frac{e^{kx}}{k} + C$	Exponential Base
Tangent Integral	$\int \tan x dx = \ln\|\sec x\| + C = -\ln\|\cos x\| + C$	Trig Standard
Secant Integral	$\int \sec x dx = \ln\|\sec x + \tan x\| + C$	Trig Standard
Inverse Sine Form	$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$	Radical Denominator
Inverse Tan Form	$\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$	Quadratic Sum
Radical Quadratic	$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln\|x + \sqrt{x^2 \pm a^2}\| + C$	Log Radical
Master Product Radical	$\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$	AOI Triad Form
Exponential Shortcut	$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$	Product Collapse
By Parts Formula	$\int uv' dx = uv - \int u'v dx$ (ILATE priority for u)	Product Rule
King's Property	$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$	Symmetry Master
Queen's Property	$\int_0^{2a} f(x) dx = 2\int_0^a f(x) dx$ if $f(2a - x) = f(x)$	Upper Limit Halver
Odd Function Rule	$\int_{-a}^{a} f(x) dx = 0$ if $f(-x) = -f(x)$	Symmetric Limits
Even Function Rule	$\int_{-a}^{a} f(x) dx = 2\int_0^a f(x) dx$ if $f(-x) = f(x)$	Symmetric Limits
Periodic Property	$\int_0^{nT} f(x) dx = n\int_0^T f(x) dx$ where $f(x + T) = f(x)$	Periodic Cycle
Leibniz Rule	$\frac{d}{dx}\left[\int_{u(x)}^{v(x)} f(t) dt\right] = f(v(x))v'(x) - f(u(x))u'(x)$	Integral Derivative
Orthogonal Parabolas	Area between $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16}{3}ab$	AOI Shortcut
Parabola and Line	Area between $y^2 = 4ax$ and $y = mx$ is $\frac{8a^2}{3m^3}$	AOI Shortcut
Standard Ellipse Area	Area of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab	AOI Shortcut

8. NTA Common Traps & High-Risk Fallacies

MASSIVE NTA TRAP WARNINGS IN INTEGRATION

Common Student Error	Mathematically Rigorous Reality	The Exact Trap Mechanism
Adding $\frac{1}{a}$ to Inverse Sine Integrals	$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$. There is NO $\frac{1}{a}$ outside!	Students confuse $\sin^{-1}\left(\frac{x}{a}\right)$ with $\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$, selecting distractor options with $\frac{1}{a}$.
Integrating Modulus Without Splitting	$\int_a^b \lvert x-c\rvert\, dx$ MUST be partitioned at root $x=c$: $\int_a^c (c-x)dx + \int_c^b (x-c)dx$.	Students drop the modulus sign directly, calculating $\frac{b^2-a^2}{2} - c(b-a)$, producing incorrect values.
Allowing Area Cancellation Below Axis	Area is strictly positive: Area $= \int_a^c f(x)dx + \lvert\int_c^b f(x)dx\rvert$.	Evaluating $\int_0^{2\pi} \sin x dx = 0$ instead of 4 sq units for total bounded area.
Applying Partial Fractions on Improper Forms	If $\deg(P) \geq \deg(Q)$, polynomial long division is MANDATORY before decomposition.	Writing $\frac{x^2}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1}$, leading to unsolvable or incorrect constants.
Forgetting Chain Factor in $e^{kx}[kf(x)+f'(x)]$	$\int e^{2x}\left[2\tan x + \sec^2 x\right]dx = e^{2x}\tan x + C$. The derivative must match $kf(x)$.	Students omit the factor 2 or divide when the inner derivative already accounts for the chain rule.
Incorrect Variable Substitution Limits	When transforming $x \to t$, integration limits MUST change: $t_1 = g(a)$ and $t_2 = g(b)$.	Students evaluate the transformed antiderivative using the original x-limits a and b.
Misidentifying Parabola Line Intersection	In $y^2 = 4ax$ and $y = mx$, limits are $x = 0$ to $x = \frac{4a}{m^2}$.	Using y-limits as x-limits or vice versa during integration strip formulation.

Practice Questions — Part 1 (Q1 to Q5)

Q1. The indefinite integral $\int \frac{\cos x - \sin x}{1+\sin 2x}dx$ is equal to:

(A) $\frac{1}{\sin x+\cos x}+C$ **(B)** $-\frac{1}{\sin x+\cos x}+C$
(C) $\ln|\sin x+\cos x|+C$ **(D)** $-\ln|\sin x+\cos x|+C$

Sol. Q1. Express the denominator using the trigonometric identity $1+\sin 2x = \cos^2 x+\sin^2 x+2\sin x\cos x = (\sin x+\cos x)^2$. The integral becomes:

$$I = \int \frac{\cos x - \sin x}{(\sin x+\cos x)^2}dx$$

Substitute $t = \sin x+\cos x \Rightarrow dt = (\cos x-\sin x)dx$:

$$I = \int \frac{dt}{t^2} = \int t^{-2}dt = -\frac{1}{t}+C = -\frac{1}{\sin x+\cos x}+C$$

Answer: (B)

Q2. The value of the integral $\int e^x\left(\frac{1+\sin x}{1+\cos x}\right)dx$ is equal to:

(A) $e^x\tan\left(\frac{x}{2}\right)+C$ **(B)** $e^x\cot\left(\frac{x}{2}\right)+C$
(C) $\frac{1}{2}e^x\tan\left(\frac{x}{2}\right)+C$ **(D)** $e^x\sec^2\left(\frac{x}{2}\right)+C$

Sol. Q2. Use half-angle formulas: $1+\sin x = 1+2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)$ and $1+\cos x = 2\cos^2\left(\frac{x}{2}\right)$. Split the fraction:

$$\frac{1+\sin x}{1+\cos x} = \frac{1}{2\cos^2\left(\frac{x}{2}\right)}+\frac{2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right)} = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)+\tan\left(\frac{x}{2}\right)$$

Let $f(x) = \tan\left(\frac{x}{2}\right) \Rightarrow f'(x) = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)$. The integral matches the master form $\int e^x[f(x)+f'(x)]dx = e^x f(x)+C$:

$$I = \int e^x\left[\tan\left(\frac{x}{2}\right)+\frac{1}{2}\sec^2\left(\frac{x}{2}\right)\right]dx = e^x\tan\left(\frac{x}{2}\right)+C$$

Answer: (A)

Q3. The value of the indefinite integral $\int \frac{dx}{x(x^7+1)}$ is equal to:

(A) $\frac{1}{7}\ln\left|\frac{x^7+1}{x^7}\right|+C$ **(B)** $\frac{1}{7}\ln\left|\frac{x^7}{x^7+1}\right|+C$
(C) $7\ln\left|\frac{x^7}{x^7+1}\right|+C$ **(D)** $\ln\left|\frac{x^7}{x^7+1}\right|+C$

Sol. Q3. Multiply numerator and denominator by x^6:

$$I = \int \frac{x^6dx}{x^7(x^7+1)}$$

Substitute $t = x^7 \Rightarrow dt = 7x^6dx \Rightarrow x^6dx = \frac{dt}{7}$:

$$I = \frac{1}{7}\int \frac{dt}{t(t+1)} = \frac{1}{7}\int\left[\frac{1}{t}-\frac{1}{t+1}\right]dt = \frac{1}{7}[\ln|t|-\ln|t+1]]+C = \frac{1}{7}\ln\left|\frac{x^7}{x^7+1}\right|+C$$

Answer: (B)

Q4. The integral $\int \frac{dx}{\sqrt{9-25x^2}}$ is equal to:

(A) $\frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right)+C$ **(B)** $\frac{1}{3}\sin^{-1}\left(\frac{5x}{3}\right)+C$
(C) $\frac{1}{15}\sin^{-1}\left(\frac{5x}{3}\right)+C$ **(D)** $\sin^{-1}\left(\frac{5x}{3}\right)+C$

Sol. Q4. Factor out 25 from the square root in the denominator:

$$\sqrt{9-25x^2} = 5\sqrt{\left(\frac{3}{5}\right)^2-x^2}$$

The integral becomes:

$$I = \frac{1}{5}\int \frac{dx}{\sqrt{\left(\frac{3}{5}\right)^2-x^2}}$$

Using the standard formula $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)+C$ with $a = \frac{3}{5}$:

$$I = \frac{1}{5}\sin^{-1}\left(\frac{x}{\frac{3}{5}}\right)+C = \frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right)+C$$

Answer: (A)

Q5. The value of $\int x^2e^{x^3}dx$ is equal to:

(A) $\frac{1}{3}e^{x^3}+C$ **(B)** $3e^{x^3}+C$
(C) $\frac{1}{2}e^{x^3}+C$ **(D)** $e^{x^3}+C$

Sol. Q5. Substitute $t = x^3$. Differentiating gives $dt = 3x^2dx \Rightarrow x^2dx = \frac{dt}{3}$. Substitute into the integral:

$$I = \int e^t\left(\frac{dt}{3}\right) = \frac{1}{3}\int e^tdt = \frac{1}{3}e^t+C = \frac{1}{3}e^{x^3}+C$$

Answer: (A)

Practice Questions — Part 2 (Q6 to Q10)

Q6. The value of the definite integral $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ is equal to:

(A) π **(B)** $\frac{\pi}{2}$
(C) $\frac{\pi}{4}$ **(D)** 0

Sol. Q6. Let $I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ …(1). Apply Property P_4 ($\int_0^a f(x)dx = \int_0^a f(a-x)dx$): $I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx$ …(2). Adding (1) and (2): $2I = \int_0^{\frac{\pi}{2}} 1dx = [x]_0^{\frac{\pi}{2}} = \frac{\pi}{2} \Longrightarrow I = \frac{\pi}{4}$. **Answer: (C)**

Q7. The value of the definite integral $\int_{-1}^{2} |x^3 - x|\, dx$ is equal to:

(A) $\frac{11}{4}$ **(B)** $\frac{7}{4}$
(C) $\frac{9}{4}$ **(D)** $\frac{5}{2}$

Sol. Q7. Roots of $x^3 - x = x(x-1)(x+1) = 0$ are $x = -1, 0, 1$. Testing signs: $x^3 - x \geq 0$ on $[-1, 0]$, $x^3 - x \leq 0$ on $[0, 1]$, and $x^3 - x \geq 0$ on $[1, 2]$. By Property P_2: $I = \int_{-1}^{0}(x^3 - x)dx - \int_0^1 (x^3 - x)dx + \int_1^2 (x^3 - x)dx$. Evaluating each: $\left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{0} - \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_1^2 = \frac{1}{4} - \left(-\frac{1}{4}\right) + \left(2 - \left(-\frac{1}{4}\right)\right) = \frac{1}{4} + \frac{1}{4} + \frac{9}{4} = \frac{11}{4}$. **Answer: (A)**

Q8. The value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^3 + x\cos x + \tan^5 x + 1)dx$ is equal to:

(A) 0 **(B)** π
(C) 2π **(D)** $\frac{\pi}{2}$

Sol. Q8. The functions x^3, $x\cos x$, and $\tan^5 x$ are all odd functions because $f(-x) = -f(x)$. By Property P_7 ($\int_{-a}^{a} f(x)dx = 0$ for odd functions), all three integrals vanish identically. Only the even constant function 1 survives:

$$I = 0 + 0 + 0 + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1dx = [x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

Answer: (B)

Q9. The value of $\lim_{x\to 0} \frac{\int_0^x \sin(t^2)dt}{x^3}$ is equal to:

(A) 0 **(B)** $\frac{1}{3}$
(C) 1 **(D)** 3

Sol. Q9. Substitution gives indeterminate form $\frac{0}{0}$. Apply L'Hôpital's Rule. Differentiating numerator via Leibniz Rule: $\frac{d}{dx}\left[\int_0^x \sin(t^2)dt\right] = \sin(x^2) \cdot 1 - 0 = \sin(x^2)$. Differentiating denominator: $\frac{d}{dx}[x^3] = 3x^2$. Evaluating limit: $\lim_{x\to 0} \frac{\sin(x^2)}{3x^2} = \frac{1}{3}\lim_{x\to 0} \frac{\sin(x^2)}{x^2} = \frac{1}{3} \cdot 1 = \frac{1}{3}$. **Answer: (B)**

Q10. The value of the definite integral $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$ is equal to:

(A) $\frac{\pi^2}{4}$ **(B)** $\frac{\pi^2}{2}$
(C) π^2 **(D)** $\frac{\pi}{4}$

Sol. Q10. Let $I = \int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$ …(1). Apply Property P_4 ($x \to \pi - x$): $I = \int_0^{\pi} \frac{(\pi-x)\sin x}{1+\cos^2 x}dx$ …(2). Adding (1) and (2): $2I = \pi\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx$. Substitute $u = \cos x \Longrightarrow du = -\sin x dx$: $2I = \pi\int_{-1}^{1} \frac{du}{1+u^2} = \pi[\tan^{-1} u]_{-1}^{1} = \pi\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = \frac{\pi^2}{2} \Longrightarrow I = \frac{\pi^2}{4}$. **Answer: (A)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. The area of the region bounded by the parabola $y^2 = 8x$ and the line $y = 2x$ is equal to:

(A) $\frac{4}{3}$ sq units (B) $\frac{8}{3}$ sq units
(C) $\frac{16}{3}$ sq units (D) $\frac{2}{3}$ sq units

Sol. Q11. Method 1 (CUET Shortcut): For parabola $y^2 = 4ax$ and line $y = mx$, Area $= \frac{8a^2}{3m^3}$. Here $4a = 8 \Longrightarrow a = 2$, and $m = 2$.

$$\text{Area} = \frac{8(2)^2}{3(2)^3} = \frac{8 \cdot 4}{3 \cdot 8} = \frac{4}{3} \text{ sq units}$$

Method 2 (Definite Integration): Find intersection points by equating: $(2x)^2 = 8x \Longrightarrow 4x^2 = 8x \Longrightarrow 4x(x-2) = 0 \Longrightarrow x = 0, 2$. Upper curve is parabola $y = \sqrt{8x} = 2\sqrt{2}x^{\frac{1}{2}}$, lower curve is line $y = 2x$:

$$\text{Area} = \int_0^2 \left[2\sqrt{2}x^{\frac{1}{2}} - 2x\right]dx = \left[2\sqrt{2} \cdot \frac{2}{3}x^{\frac{3}{2}} - x^2\right]_0^2 = \frac{4\sqrt{2}}{3}\left(2\sqrt{2}\right) - 4 = \frac{16}{3} - 4 = \frac{4}{3} \text{ sq units}$$

Answer: (A)

Q12. The area enclosed between the two parabolas $y^2 = 4x$ and $x^2 = 4y$ is equal to:

(A) $\frac{16}{3}$ sq units (B) $\frac{8}{3}$ sq units
(C) 16 sq units (D) $\frac{32}{3}$ sq units

Sol. Q12. Using the standard shortcut for orthogonal parabolas $y^2 = 4ax$ and $x^2 = 4by$:

$$\text{Area} = \frac{16ab}{3}$$

Here $4a = 4 \Longrightarrow a = 1$, and $4b = 4 \Longrightarrow b = 1$.

$$\text{Area} = \frac{16(1)(1)}{3} = \frac{16}{3} \text{ sq units}$$

Verification: Intersection is $(0,0)$ and $(4,4)$. Area $= \int_0^4 \left[2\sqrt{x} - \frac{x^2}{4}\right]dx = \left[\frac{4}{3}x^{\frac{3}{2}} - \frac{x^3}{12}\right]_0^4 = \frac{4}{3}(8) - \frac{64}{12} = \frac{32}{3} - \frac{16}{3} = \frac{16}{3}$. **Answer: (A)**

Q13. The area of the region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ in the first quadrant is:

(A) 12π sq units (B) 6π sq units
(C) 3π sq units (D) 4π sq units

Sol. Q13. For standard ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, semi-major axis $a = \sqrt{16} = 4$ and semi-minor axis $b = \sqrt{9} = 3$. Total area of the entire ellipse is given by:

$$\text{Total Area} = \pi ab = \pi(4)(3) = 12\pi \text{ sq units}$$

The first quadrant represents exactly one-fourth $(\frac{1}{4})$ of the symmetrical ellipse:

$$\text{Area}_{Q1} = \frac{1}{4}(\pi ab) = \frac{1}{4}(12\pi) = 3\pi \text{ sq units}$$

Answer: (C)

Q14. The area bounded by the curve $y = \cos x$ between $x = 0$ and $x = 2\pi$ and the x-axis is equal to:

(A) 0 sq units (B) 2 sq units
(C) 4 sq units (D) 8 sq units

Sol. Q14. The curve $y = \cos x$ crosses the x-axis at $x = \frac{\pi}{2}$ and $x = 3\frac{\pi}{2}$. Bounded area requires taking the absolute magnitude of each lobe:

- From 0 to $\frac{\pi}{2}$: $\cos x \geq 0 \Longrightarrow A_1 = \int_0^{\frac{\pi}{2}} \cos x dx = [\sin x]_0^{\frac{\pi}{2}} = 1 - 0 = 1$.
- From $\frac{\pi}{2}$ to $3\frac{\pi}{2}$: $\cos x \leq 0 \Longrightarrow A_2 = |\int_{\frac{\pi}{2}}^{3\frac{\pi}{2}} \cos x dx| = |[\sin x]_{\frac{\pi}{2}}^{3\frac{\pi}{2}}| = |-1-1| = 2$.
- From $3\frac{\pi}{2}$ to 2π: $\cos x \geq 0 \Longrightarrow A_3 = \int_{3\frac{\pi}{2}}^{2\pi} \cos x dx = [\sin x]_{3\frac{\pi}{2}}^{2\pi} = 0 - (-1) = 1$.

Total bounded area $= A_1 + A_2 + A_3 = 1 + 2 + 1 = 4$ sq units. (Note: Simple integration without modulus yields 0, which is incorrect).
Answer: (C)

Q15. The area of the region bounded by the curve $y = |x - 1|$ and the line $y = 3$ is equal to:

(A) 9 sq units (B) 6 sq units
(C) 12 sq units (D) 4.5 sq units

Sol. Q15. Find intersection points of $y = |x-1|$ and $y = 3$:

$$|x - 1| = 3 \Longrightarrow x - 1 = 3 \text{ or } x - 1 = -3 \Longrightarrow x = 4 \text{ or } x = -2$$

The bounded region is a triangle with:

- Horizontal base along $y = 3$ from $x = -2$ to $x = 4$: Base length $= 4 - (-2) = 6$.
- Vertex at the point $(1, 0)$ where the modulus reaches zero.
- Vertical height from $y = 0$ to $y = 3$: Height $= 3 - 0 = 3$.

Using the geometric formula for the area of a triangle:

$$\text{Area} = \frac{1}{2} \cdot \text{Base} \cdot \text{Height} = \frac{1}{2}(6)(3) = 9 \text{ sq units}$$

Integration verification: $\int_{-2}^{4}[3 - |x-1|]dx = \int_{-2}^{1}[3-(1-x)]dx + \int_1^4[3-(x-1)]dx = \left[2x + \frac{x^2}{2}\right]_{-2}^{1} + \left[4x - \frac{x^2}{2}\right]_1^4 = 4.5 + 4.5 = 9$. **Answer: (A)**